

WAINFLEET, United Kingdom

WMO: 033920

Lat: 53.0883N

Lon: 0.2708E

Elev: 5

StdP: 101.26

Time Zone: 0.00 (EUW)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.2	-1.1	-4.9	2.5	-0.6	-3.5	2.8	0.3	16.0	7.7	14.6	8.0	3.9	270	0.699

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.6	24.2	18.7	22.7	17.8	21.4	17.1	19.6	22.9	18.6	21.6	17.8	20.5	5.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.2	21.1	17.4	12.5	20.1	16.6	11.9	19.3	55.9	22.9	52.8	21.6	50.2	20.6	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.6	12.1	10.9	DB	-5.7	28.0	3.2	1.5	-8.0	29.0	-9.9	29.9	-11.7	30.7	-14.0	31.8	
			WB	-6.0	22.3	3.2	1.9	-8.3	23.7	-10.2	24.8	-12.0	25.9	-14.4	27.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	4.4	4.9	6.4	8.4	11.3	14.1	16.1	16.5	14.2	11.1	7.1	4.8
	DBStd	5.10	2.70	2.86	2.69	2.65	2.63	2.57	2.34	2.41	2.43	2.78	2.86	3.12
	HDD10.0	791	175	144	116	62	16	1	0	0	1	20	93	162
	HDD18.3	3074	433	376	371	298	217	129	77	67	127	223	337	419
	CDD10.0	779	0	1	3	14	57	124	189	201	126	55	7	1
	CDD18.3	22	0	0	0	0	0	2	8	10	2	0	0	0
	CDH23.3	92	0	0	0	0	2	16	32	36	5	1	0	0
	CDH26.7	7	0	0	0	0	0	2	2	3	0	0	0	0
Wind	WSAvg	5.6	6.3	6.3	5.9	5.7	5.5	4.9	4.6	4.8	5.2	5.9	6.0	6.1
Precipitation	PrecAvg	584	54	37	45	41	42	49	46	58	50	55	60	59
	PrecMax	778	108	118	115	99	95	122	137	140	120	131	114	146
	PrecMin	381	11	4	9	10	5	5	7	2	6	5	25	17
	PrecStd	102	24	24	24	21	21	30	29	31	31	35	25	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.4	13.5	15.4	19.0	21.7	25.7	26.4	26.4	24.0	19.8	15.4	12.9
		MCWB	11.1	11.5	11.8	14.4	17.1	19.7	19.7	19.9	18.8	16.1	14.0	11.5
	2%	DB	10.7	11.7	13.6	16.4	19.4	22.3	23.9	24.1	21.7	17.5	13.7	11.5
		MCWB	9.6	10.0	10.7	12.7	15.4	17.7	18.2	18.7	17.5	15.1	12.5	10.4
	5%	DB	9.6	10.4	12.1	14.6	17.6	20.4	22.3	22.6	19.8	16.2	12.5	10.5
		MCWB	8.6	8.8	9.8	11.4	14.0	16.4	17.7	17.7	16.2	14.3	11.4	9.6
	10%	DB	8.4	9.2	10.8	13.1	16.1	18.8	21.0	21.2	18.4	15.1	11.4	9.4
		MCWB	7.4	7.9	8.8	10.4	13.0	15.4	17.0	17.0	15.4	13.5	10.4	8.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	11.8	12.6	15.5	18.1	20.5	20.7	21.2	19.6	16.9	14.2	11.8
		MCDB	12.3	13.0	14.5	17.9	20.7	24.5	24.6	24.8	22.6	19.3	15.0	12.5
	2%	WB	9.9	10.3	11.3	13.4	15.9	18.4	19.3	19.5	18.0	15.7	12.7	10.7
		MCDB	10.6	11.5	13.0	15.6	18.6	21.4	22.7	22.8	20.9	17.0	13.5	11.4
	5%	WB	8.7	9.1	10.1	11.9	14.6	17.0	18.4	18.6	17.0	14.8	11.6	9.7
		MCDB	9.4	10.1	11.8	13.9	16.9	19.5	21.4	21.5	19.1	15.9	12.3	10.4
	10%	WB	7.6	8.1	9.1	10.8	13.6	16.0	17.5	17.8	16.1	13.9	10.5	8.6
		MCDB	8.3	9.1	10.5	12.7	15.5	18.3	20.3	20.3	17.8	14.9	11.2	9.4
Mean Daily Temperature Range	5% DB	MDBR	4.5	5.4	6.7	7.5	7.7	7.8	8.2	8.6	7.9	6.3	5.4	4.5
		MCDBR	6.0	6.6	8.7	9.8	10.0	10.0	10.3	10.4	9.5	7.0	5.8	5.2
		MCWBR	5.6	5.2	6.2	6.5	6.4	6.1	5.7	5.5	5.9	5.1	5.1	4.8
	5% WB	MCDBR	5.8	6.3	8.1	9.2	9.1	9.2	9.5	9.1	8.7	6.5	5.5	5.1
		MCWBR	5.5	5.3	6.2	6.6	6.4	6.2	5.9	5.4	6.0	5.2	5.1	4.8
Clear-Sky Solar Irradiance	taub	0.328	0.343	0.376	0.402	0.402	0.409	0.410	0.398	0.377	0.357	0.330	0.316	
	taud	2.353	2.384	2.293	2.238	2.291	2.294	2.302	2.355	2.411	2.442	2.405	2.335	
	Ebn at Noon	645	758	802	827	847	843	834	824	797	733	640	582	
	Edh at Noon	60	78	103	123	123	124	121	109	92	73	56	51	
All-Sky Solar Radiation	RadAvg	0.68	1.37	2.54	3.99	4.96	5.18	4.99	4.21	3.08	1.75	0.85	0.54	
	RadStd	0.07	0.19	0.25	0.41	0.37	0.40	0.47	0.29	0.26	0.14	0.09	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	+0.46	N/A	N/A	N/A	N/A	N/A
(47)	Regional (73 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page